Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

Contents

Tuesday 12 May 2026

1.1 Daily Limit

Limit Involving Exponential and Power Functions at e

Evaluate the limit:

$$\lim_{x\to e}\frac{e^x - x^e}{e - x}$$

Solution: Direct substitution of $x = e$ into the expression yields the indeterminate form $\frac{e^e - e^e}{e-e} = \frac{0}{0}$.

To resolve this indeterminacy elegantly, we can interpret the limit as the definition of a derivative or apply L'Hôpital's Rule. Let us define the differentiable function $f(x)$ in the neighborhood of $x = e$ as:

$$f(x) = e^x - x^e$$

Notice that evaluating this function at the point $x = e$ gives:

$$f(e) = e^e - e^e = 0$$

We can rewrite the given limit expression algebraically to match the standard difference quotient definition of the derivative:

$$L = \lim_{x\to e}\frac{e^x - x^e}{e - x} = -\lim_{x\to e}\frac{f(x) - f(e)}{x - e} = -f'(e)$$

Next, we differentiate $f(x)$ with respect to x using the standard exponential and power rules:

$$f'(x) = \frac{d}{dx}(e^x) - \frac{d}{dx}(x^e) = e^x - e\cdot x^{e-1}$$

Now, we evaluate the derivative at the exact point $x = e$:

$$f'(e) = e^e - e\cdot e^{e-1} = e^e - e^{1+(e-1)} = e^e - e^e = 0$$

Substituting this result back into our difference quotient relationship:

$$L = -f'(e) = -0 = 0$$

Thus, the limit evaluates strictly to zero:

$$0$$

Wednesday 13 May 2026

2.1 Daily Limit

Exponential Limit of an Average of Harmonic Numbers

Evaluate the limit:
$$\lim_{n\to\infty}\left(1+\frac{1+\frac{1}{2}+\cdots+\frac{1}{n}}{n^2}\right)^n$$

Solution: Let $H_n = \sum_{k=1}^{n}\frac{1}{k} = 1+\frac{1}{2}+\cdots+\frac{1}{n}$ denote the n-th harmonic number. The limit can then be concisely rewritten as:

$$L = \lim_{n\to\infty}\left(1+\frac{H_n}{n^2}\right)^n$$

As $n\to\infty$, we know from asymptotic analysis that the harmonic number grows logarithmically:
$$H_n = \ln(n)+\gamma+O\left(\frac{1}{n}\right)$$

where $\gamma \approx 0.5772$ is the Euler-Mascheroni constant. Therefore, the term $\frac{H_n}{n^2} \sim \frac{\ln(n)}{n^2} \to 0$, presenting us with the classical indeterminate form 1^∞.

To evaluate this systematically, we take the natural logarithm of both sides:

$$\ln L = \lim_{n\to\infty} n\ln\left(1+\frac{H_n}{n^2}\right)$$

Using the first-order Maclaurin series approximation $\ln(1+u) = u - O(u^2)$ as $u\to 0$, where $u = \frac{H_n}{n^2}$:
$$\ln\left(1+\frac{H_n}{n^2}\right) = \frac{H_n}{n^2} - O\left(\frac{H_n^2}{n^4}\right)$$

Multiplying through by n:

$$n\ln\left(1+\frac{H_n}{n^2}\right) = \frac{H_n}{n} - O\left(\frac{H_n^2}{n^3}\right)$$

Now, we take the limit as $n\to\infty$ using the logarithmic growth behavior of H_n:

$$\lim_{n\to\infty}\frac{H_n}{n} = \lim_{n\to\infty}\frac{\ln(n)+\gamma}{n} = 0$$

Since the higher-order error term $\frac{H_n^2}{n^3} \sim \frac{\ln^2(n)}{n^3}$ vanishes even faster, we conclude:
$$\ln L = 0$$

Exponentiating both sides yields the final limit value:

$$L = e^0 = 1$$

Thursday 14 May 2026

3.1 Daily Limit

Infinite Series with Quartic Denominator

Evaluate the limit:

$$\lim_{k\to\infty}\sum_{r=1}^{k}\frac{4r^2-2}{4r^4+1}$$

Solution: We begin by algebraically factoring the denominator using Sophie Germain's Identity, which states that $a^4+4b^4=(a^2-2ab+2b^2)(a^2+2ab+2b^2)$. Setting $a=1$ and $b=r$:

$$4r^4+1=(2r^2+1)^2-(2r)^2=(2r^2-2r+1)(2r^2+2r+1)$$

We now perform a partial fraction decomposition on the general term $a_r=\frac{4r^2-2}{4r^4+1}$. We seek constants A,B,C,D such that:

$$\frac{4r^2-2}{(2r^2-2r+1)(2r^2+2r+1)}=\frac{Ar+B}{2r^2-2r+1}+\frac{Cr+D}{2r^2+2r+1}$$

Clearing the denominators and equating identical polynomial terms yields:

$$4r^2-2=(Ar+B)(2r^2+2r+1)+(Cr+D)(2r^2-2r+1)$$

By inspection or system solving, we find that setting $A=2$, $B=-1$, $C=-2$, and $D=-1$ satisfies the identity:

$$\frac{4r^2-2}{4r^4+1}=\frac{2r-1}{2r^2-2r+1}-\frac{2r+1}{2r^2+2r+1}$$

Let us define the rational function $f(r)=\frac{2r-1}{2r^2-2r+1}$. Observe what happens when we evaluate $f(r+1)$:

$$f(r+1)=\frac{2(r+1)-1}{2(r+1)^2-2(r+1)+1}=\frac{2r+1}{(2r^2+4r+2)-(2r+2)+1}=\frac{2r+1}{2r^2+2r+1}$$

This reveals that our general term is precisely a difference of consecutive terms:

$$a_r=f(r)-f(r+1)$$

Summing this from $r=1$ to k creates a telescoping series where all intermediate terms cancel out:

$$S_k=\sum_{r=1}^{k}[f(r)-f(r+1)]=f(1)-f(k+1)$$

We evaluate the initial boundary term at $r=1$:

$$f(1)=\frac{2(1)-1}{2(1)^2-2(1)+1}=\frac{1}{1}=1$$

Taking the limit as $k\to\infty$ for the remaining boundary term:

$$\lim_{k\to\infty}f(k+1)=\lim_{k\to\infty}\frac{2k+1}{2k^2+2k+1}=0$$

Therefore, the infinite sum converges cleanly to:

$$\lim_{k\to\infty}S_k=1-0=1$$

Friday 15 May 2026

4.1 Daily Limit

Limit Involving Trigonometric, Logarithmic, and Exponential Functions

Evaluate the limit:

$$\lim_{x\to 0}\left(\frac{\tan(x^2)-\ln(\cos(x))}{x^2e^x}\right)$$

Solution: Direct substitution of $x=0$ gives the indeterminate form $\frac{0-0}{0\cdot 1}=\frac{0}{0}$. We evaluate this limit using standard Maclaurin series expansions around $x=0$.

First, recall the expansion for the tangent function applied to $u=x^2$:

$$\tan(x^2)=x^2+\frac{1}{3}(x^2)^3+O(x^8)=x^2+O(x^6)$$

Second, we derive the series expansion for $\ln(\cos(x))$. Using the cosine series $\cos(x)=1-\frac{x^2}{2}+\frac{x^4}{24}-O(x^6)$:

$$\ln(\cos(x))=\ln\left(1-\left(\frac{x^2}{2}-\frac{x^4}{24}+O(x^6)\right)\right)$$

Using the logarithmic expansion $\ln(1-u)=-u-\frac{u^2}{2}-O(u^3)$:

$$\ln(\cos(x))=-\left(\frac{x^2}{2}-\frac{x^4}{24}\right)-\frac{1}{2}\left(\frac{x^2}{2}\right)^2-O(x^6)$$

$$=-\frac{x^2}{2}+\frac{x^4}{24}-\frac{x^4}{8}-O(x^6)=-\frac{x^2}{2}-\frac{x^4}{12}-O(x^6)$$

Now, we subtract the logarithmic expansion from the tangent expansion in the numerator:

$$\tan(x^2)-\ln(\cos(x))=[x^2+O(x^6)]-\left[-\frac{x^2}{2}-\frac{x^4}{12}-O(x^6)\right]=\frac{3}{2}x^2+\frac{1}{12}x^4+O(x^6)$$

For the denominator, we expand $e^x=1+x+O(x^2)$:

$$x^2e^x=x^2\left(1+x+O(x^2)\right)=x^2+x^3+O(x^4)$$

Substituting our algebraic approximations back into the original ratio:

$$L=\lim_{x\to 0}\frac{\frac{3}{2}x^2+\frac{1}{12}x^4+O(x^6)}{x^2+x^3+O(x^4)}$$

Factoring out the dominant power x^2 from both numerator and denominator:

$$L=\lim_{x\to 0}\frac{x^2\left(\frac{3}{2}+\frac{1}{12}x^2+O(x^4)\right)}{x^2\left(1+x+O(x^2)\right)}=\lim_{x\to 0}\frac{\frac{3}{2}+\frac{1}{12}x^2+O(x^4)}{1+x+O(x^2)}$$

Evaluating the remaining expression at $x=0$ removes all indeterminacy:

$$L=\frac{\frac{3}{2}+0}{1+0}=\frac{3}{2}$$

Saturday 16 May 2026

5.1 Daily Limit

Limit of a Riemann Sum Involving the Floor Function

Evaluate the limit:

$$\lim_{n\to\infty}\sum_{k=1}^{n}\frac{\lfloor\sqrt{n^2+kn}\rfloor}{n^2}$$

Solution: We evaluate this limit using the Squeeze (Sandwich) Theorem. By the fundamental definition of the floor function, for any real number y, we have the strict algebraic inequalities:

$$y-1<\lfloor y\rfloor\le y$$

Setting $y=\sqrt{n^2+kn}$ and dividing all terms of the inequality by n^2 gives:

$$\frac{\sqrt{n^2+kn}-1}{n^2}<\frac{\lfloor\sqrt{n^2+kn}\rfloor}{n^2}\le\frac{\sqrt{n^2+kn}}{n^2}$$

Summing these expressions over the index k from 1 to n:

$$\sum_{k=1}^{n}\frac{\sqrt{n^2+kn}}{n^2}-\sum_{k=1}^{n}\frac{1}{n^2}<\sum_{k=1}^{n}\frac{\lfloor\sqrt{n^2+kn}\rfloor}{n^2}\le\sum_{k=1}^{n}\frac{\sqrt{n^2+kn}}{n^2}$$

Notice that the subtracted summation term is a simple constant sum:

$$\sum_{k=1}^{n}\frac{1}{n^2}=n\cdot\frac{1}{n^2}=\frac{1}{n}$$

As $n\to\infty$, this difference $\frac{1}{n}\to 0$. Therefore, both the lower and upper bounds share the exact same asymptotic limit. We can focus solely on evaluating the continuous sum:

$$S=\lim_{n\to\infty}\sum_{k=1}^{n}\frac{\sqrt{n^2+kn}}{n^2}$$

We factor out n^2 from under the radical:

$$S=\lim_{n\to\infty}\sum_{k=1}^{n}\frac{\sqrt{n^2\left(1+\frac{k}{n}\right)}}{n^2}=\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}\sqrt{1+\frac{k}{n}}$$

This expression is a right-endpoint Riemann sum for the continuous function $f(x)=\sqrt{1+x}$ over the interval $[0,1]$, where $\Delta x=\frac{1}{n}$ and $x_k=\frac{k}{n}$. Passing to the definite integral:

$$S=\int_0^1\sqrt{1+x}\,dx$$

Using the power rule for integration $\int u^{1/2}du=\frac{2}{3}u^{3/2}$:

$$S=\left[\frac{2}{3}(1+x)^{3/2}\right]_0^1=\frac{2}{3}\left((1+1)^{3/2}-(1+0)^{3/2}\right)=\frac{2}{3}\left(2^{3/2}-1\right)=\frac{2}{3}\left(2\sqrt{2}-1\right)$$

By the Squeeze Theorem, the original floor sum converges strictly to this same value:

$$\frac{4\sqrt{2}-2}{3}$$

Sunday 17 May 2026

6.1 Daily Limit

Trigonometric Exponential Indeterminate Limit

Evaluate the limit:

$$\lim_{x\to 0}\left(\frac{\sin(x)}{x}\right)^{\frac{1}{1-\cos(x)}}$$

Solution: As $x \to 0$, we have $\frac{\sin(x)}{x} \to 1$ and $1-\cos(x) \to 0$, producing the classic indeterminate exponential form 1^∞. To evaluate this systematically, we take the natural logarithm:

$$\ln L = \lim_{x\to 0}\frac{1}{1-\cos(x)}\ln\left(\frac{\sin(x)}{x}\right) = \lim_{x\to 0}\frac{\ln\left(\frac{\sin(x)}{x}\right)}{1-\cos(x)}$$

We expand both the numerator and the denominator using their Maclaurin series around $x = 0$. For the denominator:

$$1-\cos(x) = 1-\left(1-\frac{x^2}{2}+\frac{x^4}{24}-O(x^6)\right) = \frac{x^2}{2}-\frac{x^4}{24}+O(x^6) \approx \frac{x^2}{2}$$

For the numerator, we first expand $\frac{\sin(x)}{x}$:

$$\frac{\sin(x)}{x} = \frac{x-\frac{x^3}{6}+\frac{x^5}{120}-O(x^7)}{x} = 1-\frac{x^2}{6}+\frac{x^4}{120}-O(x^6)$$

Now, substitute this into the Taylor expansion $\ln(1-u) = -u - O(u^2)$ where $u = \frac{x^2}{6}-\frac{x^4}{120}$:

$$\ln\left(\frac{\sin(x)}{x}\right) = \ln\left(1-\left(\frac{x^2}{6}-\frac{x^4}{120}\right)\right) = -\frac{x^2}{6}+O(x^4)$$

Substituting both leading asymptotic terms back into our logarithmic limit ratio:

$$\ln L = \lim_{x\to 0}\frac{-\frac{x^2}{6}+O(x^4)}{\frac{x^2}{2}-O(x^4)}$$

Dividing both numerator and denominator by x^2:

$$\ln L = \lim_{x\to 0}\frac{-\frac{1}{6}+O(x^2)}{\frac{1}{2}-O(x^2)} = \frac{-1/6}{1/2} = -\frac{1}{3}$$

Exponentiating both sides to solve for L yields:

$$L = e^{-1/3} = \frac{1}{\sqrt[3]{e}}$$

Monday 18 May 2026

7.1 Daily Limit

Limit of a Sum of Logarithms

Evaluate the limit:

$$\lim_{n\to\infty}\sum_{k=1}^{n}\ln\left(1+\frac{k}{n^2}\right)$$

Solution: We analyze the argument of the logarithm for large n. Since the summation index k ranges from 1 to n, the fraction $u_k = \frac{k}{n^2}$ satisfies the bound:

$$0 < \frac{k}{n^2} \leq \frac{n}{n^2} = \frac{1}{n}$$

As $n \to \infty$, $u_k \to 0$ uniformly for all terms in the sum. We can therefore apply the Taylor series expansion for the logarithm: $\ln(1+u) = u - \frac{u^2}{2} + O(u^3)$:

$$\ln\left(1+\frac{k}{n^2}\right) = \frac{k}{n^2} - \frac{1}{2}\left(\frac{k}{n^2}\right)^2 + O\left(\frac{k^3}{n^6}\right) = \frac{k}{n^2} - \frac{k^2}{2n^4} + O\left(\frac{k^3}{n^6}\right)$$

Summing this algebraic representation across all k from 1 to n:

$$\sum_{k=1}^{n}\ln\left(1+\frac{k}{n^2}\right) = \sum_{k=1}^{n}\frac{k}{n^2} - \sum_{k=1}^{n}\frac{k^2}{2n^4} + \sum_{k=1}^{n}O\left(\frac{k^3}{n^6}\right)$$

We factor out powers of n from each summation and evaluate using standard polynomial formulas: 1. The primary linear term:

$$\frac{1}{n^2}\sum_{k=1}^{n}k = \frac{1}{n^2}\left[\frac{n(n+1)}{2}\right] = \frac{n^2+n}{2n^2} = \frac{1}{2} + \frac{1}{2n}$$

2. The secondary quadratic error term:

$$\frac{1}{2n^4}\sum_{k=1}^{n}k^2 = \frac{1}{2n^4}\left[\frac{n(n+1)(2n+1)}{6}\right] = \frac{2n^3+3n^2+n}{12n^4} = O\left(\frac{1}{n}\right)$$

3. The cubic remainder term is bounded by:

$$\sum_{k=1}^{n}O\left(\frac{k^3}{n^6}\right) = O\left(\frac{1}{n^6}\sum_{k=1}^{n}k^3\right) = O\left(\frac{1}{n^6}\cdot\frac{n^4}{4}\right) = O\left(\frac{1}{n^2}\right)$$

Substituting these summed results back into our expression:

$$\sum_{k=1}^{n}\ln\left(1+\frac{k}{n^2}\right) = \left(\frac{1}{2}+\frac{1}{2n}\right) - O\left(\frac{1}{n}\right) + O\left(\frac{1}{n^2}\right)$$

Taking the limit as $n \to \infty$, all inverse powers of n vanish completely:

$$\lim_{n\to\infty}\sum_{k=1}^{n}\ln\left(1+\frac{k}{n^2}\right) = \frac{1}{2}$$

Tuesday 19 May 2026

8.1 Daily Limit

Exponential Trigonometric Limit of Form 1^∞

Evaluate the limit:
$$\lim_{x\to 0}(\cos(x))^{\cot^2(x)}$$

Solution: As $x \to 0$, $\cos(x) \to 1$ and $\cot^2(x) \to \infty$, yielding the indeterminate form 1^∞. We take the natural logarithm of the limit expression:

$$\ln L = \lim_{x\to 0} \cot^2(x)\ln(\cos(x))$$

Using the trigonometric identity $\cot(x) = \frac{1}{\tan(x)}$, we can rewrite this product as a standard quotient:
$$\ln L = \lim_{x\to 0} \frac{\ln(\cos(x))}{\tan^2(x)}$$
We evaluate this limit by substituting Maclaurin series expansions for both functions around $x = 0$: 1. For the denominator, since $\tan(x) = x + \frac{x^3}{3} + O(x^5)$:

$$\tan^2(x) = (x + O(x^3))^2 = x^2 + O(x^4)$$

2. For the numerator, we first expand $\cos(x) = 1 - \frac{x^2}{2} + O(x^4)$. Substituting into $\ln(1-u) = -u - O(u^2)$ where $u = \frac{x^2}{2} - O(x^4)$:

$$\ln(\cos(x)) = \ln\left(1 - \left(\frac{x^2}{2} - O(x^4)\right)\right) = -\frac{x^2}{2} + O(x^4)$$

Substituting both expansions back into the quotient:

$$\ln L = \lim_{x\to 0} \frac{-\frac{x^2}{2} + O(x^4)}{x^2 + O(x^4)}$$

Dividing numerator and denominator by the common factor x^2:

$$\ln L = \lim_{x\to 0} \frac{-\frac{1}{2} + O(x^2)}{1 + O(x^2)} = -\frac{1}{2}$$

Exponentiating both sides yields the final value:

$$L = e^{-1/2} = \frac{1}{\sqrt{e}}$$

Wednesday 20 May 2026

9.1 Daily Limit

Limit of an Exponential Difference of Square Roots

Evaluate the limit:
$$\lim_{x\to\infty}\left[\left(\sqrt{x^2+2x-1}-\sqrt{x^2-1}\right)^x\right]$$

Solution: We first rationalize the difference of square roots inside the base by multiplying and dividing by the conjugate expression $\left(\sqrt{x^2+2x-1}+\sqrt{x^2-1}\right)$:

$$\sqrt{x^2+2x-1}-\sqrt{x^2-1}=\frac{(x^2+2x-1)-(x^2-1)}{\sqrt{x^2+2x-1}+\sqrt{x^2-1}}=\frac{2x}{\sqrt{x^2+2x-1}+\sqrt{x^2-1}}$$

Divide the numerator and denominator by x:

$$=\frac{2}{\sqrt{1+\frac{2}{x}-\frac{1}{x^2}}+\sqrt{1-\frac{1}{x^2}}}$$

We now determine the asymptotic behavior of the denominator as $x\to\infty$ using the generalized binomial expansion $(1+u)^{1/2}=1+\frac{1}{2}u-\frac{1}{8}u^2+O(u^3)$: 1. For the first radical where $u=\frac{2}{x}-\frac{1}{x^2}$:

$$\sqrt{1+\frac{2}{x}-\frac{1}{x^2}}=1+\frac{1}{2}\left(\frac{2}{x}-\frac{1}{x^2}\right)-\frac{1}{8}\left(\frac{2}{x}\right)^2+O(x^{-3})=1+\frac{1}{x}-\frac{1}{x^2}+O(x^{-3})$$

2. For the second radical where $u=-\frac{1}{x^2}$:

$$\sqrt{1-\frac{1}{x^2}}=1-\frac{1}{2x^2}+O(x^{-4})$$

Summing these two radical expansions together:

$$\text{Denominator}=2+\frac{1}{x}-\frac{3}{2x^2}+O(x^{-3})=2\left(1+\frac{1}{2x}-\frac{3}{4x^2}+O(x^{-3})\right)$$

Now, substitute this expanded denominator back into our fraction for the base:

$$\text{Base}=\frac{2}{2\left(1+\frac{1}{2x}-\frac{3}{4x^2}+O(x^{-3})\right)}=\left(1+\frac{1}{2x}-\frac{3}{4x^2}+O(x^{-3})\right)^{-1}$$

Using the geometric inversion series $(1+v)^{-1}=1-v+v^2-O(v^3)$ where $v=\frac{1}{2x}-\frac{3}{4x^2}$:

$$\text{Base}=1-\left(\frac{1}{2x}-\frac{3}{4x^2}\right)+\left(\frac{1}{2x}\right)^2+O(x^{-3})=1-\frac{1}{2x}+\frac{1}{x^2}+O(x^{-3})$$

Now, we raise this simplified base to the power x and take the natural logarithm to evaluate:

$$\ln L=\lim_{x\to\infty}x\ln\left(1-\frac{1}{2x}+\frac{1}{x^2}+O(x^{-3})\right)$$

Using $\ln(1+w) = w - O(w^2)$ where $w = -\frac{1}{2x} + \frac{1}{x^2}$:

$$\ln L = \lim_{x\to\infty} x\left(-\frac{1}{2x} + O(x^{-2})\right) = \lim_{x\to\infty}\left(-\frac{1}{2} + O(x^{-1})\right) = -\frac{1}{2}$$

Exponentiating both sides yields the final limit value:

$$L = e^{-1/2} = \frac{1}{\sqrt{e}}$$

Thursday 21 May 2026

10.1 Daily Limit

Nested Radical Limit Involving x-th Roots

Evaluate the limit:

$$\lim_{x\to 0} \sqrt[x]{\frac{\sqrt[x]{1+x}}{\sqrt[x]{\frac{1}{1-x}}}}$$

Solution: To simplify this nested radical expression, we rewrite all root operations using rational exponents. Let us denote the expression inside the limit as $E(x)$:

$$E(x) = \left(\frac{(1+x)^{1/x}}{\left(\frac{1}{1-x}\right)^{1/x}} \right)^{1/x}$$

First, simplify the fraction in the denominator using negative exponents:

$$\left(\frac{1}{1-x}\right)^{1/x} = \left((1-x)^{-1}\right)^{1/x} = (1-x)^{-1/x}$$

Substitute this back into the inner ratio:

$$\frac{(1+x)^{1/x}}{(1-x)^{-1/x}} = (1+x)^{1/x} \cdot (1-x)^{1/x}$$

Since both factors share the exact same exponent $\frac{1}{x}$, we can combine their bases using the difference of squares factorization $(a+b)(a-b) = a^2 - b^2$:

$$[(1+x)(1-x)]^{1/x} = (1-x^2)^{1/x}$$

Now, apply the outermost x-th root exponent:

$$E(x) = \left((1-x^2)^{1/x}\right)^{1/x} = (1-x^2)^{1/x^2}$$

Our limit expression is now dramatically simplified to:

$$L = \lim_{x\to 0} (1-x^2)^{\frac{1}{x^2}}$$

To make the final evaluation transparent, we introduce a change of variable by setting $u = -x^2$. As $x \to 0$, it is clear that $u \to 0^-$. Substituting this variable transformation into our limit:

$$L = \lim_{u\to 0^-} (1+u)^{-\frac{1}{u}} = \lim_{u\to 0^-} \left[(1+u)^{\frac{1}{u}}\right]^{-1}$$

We recognize the inner bracket as the fundamental limit definition of Euler's number e:

$$\lim_{u\to 0} (1+u)^{\frac{1}{u}} = e$$

Therefore, substituting this value yields:

$$L = e^{-1} = \frac{1}{e}$$